

and these have low conversion efficiencies.

These batteries would be highly useful in environments where conventional power sources cannot easily be replaced, such as implanted hearing aids, artificial pancreas, or pacemakers. No extension cord is long enough to take to another planet, and we know that there is no spacecraft charging station available during the journey as yet. That's why researchers are working hard on ways to make spacecraft power systems more efficient, resilient, and long-lasting.

Due to a very long life, diamond nuclear batteries could power spacecraft or a satellite, increasing the flight time and enabling much faraway travel than what is currently possible.

A tiny, robust power source that does not require any mediator interface, wiring, or conductors but only a chemical reaction that generates a low, continuous electrical current has immense value in outer space.

The high energy capacity of nuclear batteries makes them suitable for applications ranging from environmental health monitoring to structural health monitoring. These are also used in a variety of industrial applications, including electron capture devices for gas chromatography. The technology has already been established in practice on a small scale at the University of Bristol.

One of the challenges now is to put in place an assembly line for these diamond batteries for large scale produc-



Space exploration is a key application area of nuclear diamond batteries (Source: <https://en.wikipedia.org>)

tion as soon as possible. Another issue that needs to be tackled is regulatory acceptance and approval, that is, what is acceptable by regulatory bodies in different countries and what is not in terms of manufacturing such batteries for a mainstream consumer market.

Conclusion

The nuclear-based radioactive isotopes batteries involve no moving parts, zero maintenance, and just direct and continuous electricity generation.

The power output of these diamond batteries is low as compared to other alternatives. However, their total longevity favours them to become critical components of future electronics manufacturing. By encapsulating

radioactive materials inside diamonds, it is possible to turn a long-term serious problem of nuclear waste management into a nuclear-powered battery for a long-term supply of clean energy.

A large number of graphite blocks are stored in various parts of the world. When Britain's Magnox Nuclear reactors were withdrawn in 2015 after more than fifty years of service, the nuclear waste consisted of 95,000 tonnes of graphite blocks used as moderators to manage the nuclear fission process. This technology would make it possible to significantly reduce radioactivity and thus the cost of disposal.

Comparatively, the cost of producing an artificial diamond is much less than disposing of used nuclear fuel and nuclear waste. It is expected that by 2021, despite efforts, the US would have radioactive fuel, which is sufficient only for two and a half nuclear battery modules a year. However, the Mars rover known as 'Curiosity' itself required eight nuclear battery modules.

This scarcity and the small existing inventory are inadequate for the ambitious planetary missions planned till 2030 to destinations such as the very cold moons of Jupiter and Saturn. Hence, NASA has been examining and analysing alternatives. Continuous research and development are in progress for making safe, compact, high energy capacity, and long life batteries a reality for applications ranging from environmental, medical to structural health monitoring. **EFY**



Mars rover called 'Curiosity' required eight nuclear battery modules (Source: www.livescience.com)